7 The Simple Pendulum

Introduction

Consider a mass hanging from the ceiling by a thin string.
If we displace the mass from equilibrium and release it from rest, the mass will oscillate.
Over time, due to air resistance, the oscillation amplitude damps until eventually the system rests at the equilibrium position. Every time the pendulum is released under the same conditions, it will oscillate at the same frequency and will describe a similar path.
From our everyday *observation* one may notice that changing the length of the pendulum leads to a change in the pendulum period:

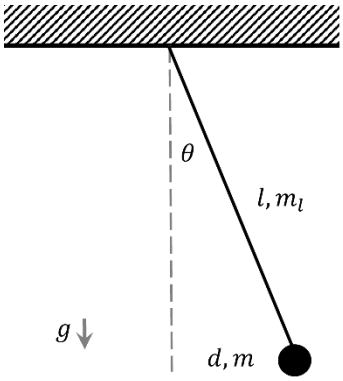


Figure 7.1 The simple pendulum

A long pendulum will oscillate slower than a short one. A pendulum which is released from a larger angle, will oscillate with large amplitude. However, replacing the mass by a heavier one leads to no noticeable change in the oscillation pace.
Many interesting questions can be investigated in the pendulum system. However, the *question* we want to answer is:
What is the mathematical relation between the pendulum period and the different physical parameters of the system?

- If we compare two identical pendulums, what is the mathematical relation between the length of the pendulum and its period?
- If we compare two identical pendulums, but only with different masses: one with lighter mass than the second one, will have different period?
- If we compare two identical pendulums, but only released from different initial angles: one released from small angle, and one from a bit larger angle, will they have different period?

Theoretical model

We first may answer our questions by deriving a ***theoretical model.*** Mechanical theoretical model consists in writing and solving the equations of motion under the appropriate assumptions for the forces in the system according to Newton's laws.
Solving the equation of motion can sometimes be somewhat challenging and could require solution of quite difficult differential equations. The derivation of the theoretical model describing the period of the pendulum can be found in the first reference at the end of this guide [1] and preparation questions 2 and 3 (It is advisable to take a look at the theoretical derivation).

However, often, it is possible to find the mathematical relation between a physical variable and a dependent variable simply by ***dimensional analysis*** without the need to solve the equations of motion. The concept of physical dimension, and of dimensional analysis, was introduced by Joseph Fourier in 1822.

First, we point out that the physical dimensions of any physical quantity can be expressed as the product of powers of the fundamental dimensions. The fundamental 3 dimensions in mechanics being mass $[M]$, length $[L]$, and time $[T]$. Hence, the dimension of any physical mechanical quantity can be expressed in the form:

$$[physical\ quantity] = [M]^{\alpha} \cdot [L]^{\beta} \cdot [T]^{\gamma} \qquad \text{(eq. 1)}$$

Where the constants α, β and γ determine in which units the physical quantity has to be measured.

Note: In our lab we will use the SI system of units exclusively. The mass is measured in kilograms. Length is meters and time in seconds.

As an example, let us consider dimensional analysis of the gravitational constant g,

$$[g] = [m/s^2] = [M]^0 \cdot [L]^1 \cdot [T]^{-2} \qquad \text{(eq. 2)}$$

Secondly, we indicate that any physical equation, or inequality, must have the same dimensions on its left and right sides, a property known as dimensional consistency [2]. The method of dimensional analysis is to list all the physical parameters of the system and using their dimensions to find the functional relation between them.

The pendulum system consists of 8 physical parameters (figure 6.1.1):

Length of the pendulum l $[m]$, thread mass m_l $[kg]$, diameter of the weight d $[m]$, hanging mass m $[kg]$, initial release angle θ_0 $[rad]$, the gravitational constant g $[m/s^2]$, air resistance (drag b) and finally the period of the motion T $[sec]$.

We aim to find how the period T depends on the other parameters:

$$T = f(l, m_l, d, m, \theta_0, g, b) \qquad \text{(eq. 3)}$$

This is a problem, because we have to many possibilities to arrange the parameters into the groups, and we cannot find the relation for the period as we aim.

Therefore, we must simplify our model by making some *assumptions*. Our assumptions must be reasonable and in accordance with our knowledge of the physical system. Hereby, **we shall assume a simple pendulum system – A point mass hanging from a very thin, stiff and massless string and we will assume negligible friction. Hence, the conditions to be satisfied by the simple pendulum system are: $\boldsymbol{d \ll l}$, $\boldsymbol{m_l \ll m_d}$ and no loss of energy ($\boldsymbol{b \to 0}$).**

According to our simplifying assumptions, the parameters of the physical system are: the length of the pendulum l $[m]$, the mass of the pendulum m $[kg]$, the gravitational acceleration g $[m/s^2]$ and the initial angular amplitude θ_0.

Our problem regarding the simple pendulum is then, to find a function f such that

$$T = f(l, m, g, \theta_0) \qquad \text{(eq. 4)}$$

Let us write the dimensions of this equation:

We begin by writing the period as a product of these given quantities that have dimensions, (thus the initial angular amplitude of the pendulum cannot enter our expression), with each given quantity raised to a rational power,

$$[T] = [l]^{\alpha}[m]^{\beta}[g]^{\gamma} \qquad \text{(eq. 5)}$$

To perform dimensional analysis of this equation, let us substitute the dimension of each physical quantity in place of the physical quantity itself

$$[T] = [L]^{\alpha}[M]^{\beta}[LT^{-2}]^{\gamma} \qquad \text{(eq. 6)}$$

Next, we arrange all the physical dimensions:

$$[T] = [L]^{\alpha+\gamma}[M]^{\beta}[T]^{-2\gamma} \qquad \text{(eq. 7)}$$

Finally, we demand equality of the dimensions on both sides of the equation. The dimensions of every physical equation must be equal from both sides of the equation! This leads to the following:

$$\alpha = 1/2, \beta = 0, \text{and } \gamma = -1/2 \qquad \text{(eq. 8)}$$

Thus, considering only the dimensions of the physical parameters of the system, we obtained the functional relation between the period of the pendulum oscillations and the length of the pendulum:

$$T(\theta_0, l) = C(\theta_0) \cdot \sqrt{\frac{l}{g}} \qquad \text{(eq. 9)}$$

where C is an unknown constant or some function of θ_0.

The dependence on l and g is just what we should expect. A long pendulum has a longer period than a shorter one. Increasing g increases the restoring force, causing the frequency to increase and the period to decrease. We learn that dimensional analysis provides an easy way to determine functional relation between the parameters of the system, up to an unknown constant.

Note, however, that since the release angle θ_0 is dimensionless, it did not appear in the dimensionless analysis. So, it is possible that the constant C is some unknown function of the releasing angle θ_0. This $C(\theta_0)$ may be identified by solving the equations of motion (if it is possible), or experimentally (by repeating the same experiment for different release angles).

Hence, our model predicts that T will vary as $\sqrt{l}$ for a constant θ_0. If we plot T as function of $\sqrt{l}$ we will expect to get a linear line with intercept zero. If we do not obtain a reasonably linear line, then we need to reexamine the assumptions regarding the experiment.

We could isolate the effect of $C(\theta_0)$ on the pendulum period by holding l constant and varying θ_0. If we plot T as function of θ_0 we may reveal the function $C(\theta_0)$.

Preparation Questions

1. A constant force of magnitude F acts on a body with mass m over a distance d.
   1.1 Write down an expression for the final velocity of the body using dimensional analysis.
   1.2 Verify your calculation is correct and find the unknown constant using work-kinetic energy theorem.
2. Planning for the experiment (experimental considerations). Write down the assumptions of our model. How can we achieve these conditions in the experiment we will perform?
   2.1 Think: how we produce a massless thread in the experiment?
   2.2 Think: how we produce a point mass in the experiment?
   2.3 Think: how we produce a one-dimension movement of the pendulum in the experiment?
   2.4 In our model we have assumed that the resistance of air is negligible. In which way non-negligible air resistance will affect the measurements?
3. Equation of motion and its relation to dimensional analysis
   3.1 Draw the diagram of forces acting on the simple pendulum according to figure 1.
   3.2 Write down the equation of motion for the simple pendulum. Can you solve it analytically?
   3.3 What are the physical parameters appearing in your equation? (you should get the same physical parameters as we got in our dimensional analysis derivation).
4. In order to solve analytically the equation of motion you got in question 3, it is necessary to approximate for small angles $\sin\theta \approx \theta$. Under this approximation, the solution of the equation of motion gives the relation between the period of the pendulum and its length: $T = 2\pi\sqrt{l\backslash g}$ (Recall equation 9 we got using dimensionless analysis!).
   4.1 What is the meaning of the "small angle approximation"? For which angles does this approximation hold? Find the angle at which the relative error of the small angle approximation exceeds 1%.
   4.2 Try to solve the equation of motion to find the motion of the pendulum $\theta(t)$.
   4.3 Try to find $T(l)$.

Experimental procedure and data analysis

Write Python script logging all the values of all measured variables including measurement errors. Write in-line comments to explain every variable and its units.

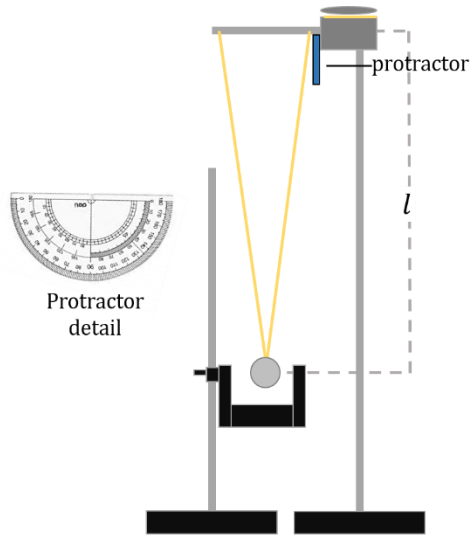


Figure 7.2 Experimental setup

Part A: Preliminary experiment

1. The mass of the pendulum is indicated on the pendulum base. Using the caliper, carefully measure the diameter of the pendulum. Estimate the error of these measurements.
2. Connect the photogate to the Airlink. Turn on the Airlink and associate it with PASCO Capstone. Then load the experiment setup file you have downloaded from Moodle.
3. Measure the period of oscillations according to the procedure explained to you by the lab tutor. Use the protractor do measure the angle of release. **Try, as possible, to release the mass from the center and at rest when the thread is thought and keep the system stable. Note that the length of the pendulum is the distance from the fixed point to the center of mass of the hanging weight.**
4. Measure the period of oscillations of the pendulum for a certain length and **a small angle** (verify that the assumptions of the theoretical model are fulfilled). **To get more accurate result, take the period as average over several oscillations of the pendulum.** Use the result of preparation question 4 in order to calculate the experimental value of the gravitational constant $g_{exp,1}$ and compare it to the known value in Shantou.

Part B: Period of the pendulum for different lengths at the same small release angle

5. **Measure the period of oscillation of the pendulum for at least 6 different lengths for the same small release angle. To get more accurate result, take the period as average over several**

oscillations of the pendulum. Choose lengths such that the assumptions of the theoretical model hold.

6. Plot the results of your measurements. Use curve fitting and goodness of fit to decide whether the theoretical model agrees with the experiment? Discuss you results.
7. **Use the results of the curve fitting to find the experimental gravitational constant $g_{exp,2}$ and compare it to the analytical model you have derived in preparation question 4 using the value in Shantou. What is your conclusion from this comparison?**

Part C: Period of the pendulum with the same length and at different angles

8. **Measure the period of oscillation of the pendulum for different initial angles from which the weight is released for the same pendulum length. Repeat this measure for at least 10 different initial angles.** (Recommended angles: $\mathbf{3^0, 6^0, 9^0, 12^0, 15^0, 20^0, 25^0, 30^0, 40^0, 50^0, 60^0}$).
9. Plot the period of oscillations as function of the initial angle of release.
10. What is the limit of small angles in your system? Do your results in the small angle regime agree with the analytical solution of the simple pendulum (preparation question 4)?
11. Find the function $C(\theta_0)$ from your measurements (you may use reference 3 equation 6 and reference 4 for assistance).

References

[1] Walker, J., Halliday, D., Resnick, R., & Trees, B. R. (2018). *Fundamentals of physics 9th edition*. Ch. 15, pp. 395-398, Wiley.

[2] Langhaar, H. L. (1951). *Dimensional Analysis and Theory of Models*. John Wiley & Sons, Inc. New York, Ch. 2, Pgs. 13-16.

[3] Beléndez, A., Hernández, A., Márquez, A., Beléndez, T., & Neipp, C. (2006). Analytical approximations for the period of a nonlinear pendulum. European Journal of Physics, 27(3), 539.

[4] Amrani, D., Paradis, P., & Beaudin, M. (2008). Approximation expressions for the large-angle period of a simple pendulum revisited. Revista mexicana de física E, 54(1), 59-64.

[5] Adams, B. (06.02.2014), Local Acceleration of Gravity, http://www.wolframalpha.com/widgets/view.jsp?id=e856809e0d522d3153e2e7e8ec263bf2